$\log_a(x) = y \Leftrightarrow a^{y=x}$

$a^x = a^y \Leftrightarrow x = y$

Products and Quotients Inside a Logarithm

For $u, v > 0$ and $a > 0$, $a \neq 1$:

② $\log_a(uv) = \log_a u + \log_a v$, ① $\log_a\left(\frac{u}{v}\right) = \log_a u - \log_a v$.

Evaluate exactly: $\log_2 4 + \log_2 8$ and $\log_5 75 - \log_5 3$.

- $\log_a(u+v) \neq \log_a u + \log_a v$

- $\log_2 4 + \log_2 8 = \log_2(32)$

  use ①

$\log_2(32) = y$

$2^y = 32$

$2^y = 2^5$

$y = 5$

- $\log_5 75 - \log_5 3 = \log_5\left(\frac{75}{3}\right)$

  ②

$= \log_5(25)$

$= \log_5(5^2) = 2$

Powers and Inverse Properties

For $u > 0$, $\log_a(u^r) = r \log_a u$. Also $\log_a 1 = 0$, $\log_a a = 1$, and $a^{\log_a u} = u$. Simplify exactly:

(a) $\ln(e^5)$ (b) $4^{\log_4 3 + \log_4 2}$ (c) $\log_7 1$.

Use $a^r = e^{r \ln a}$ to rewrite 3^{2x} using base e.

- $\log_a(u^3) = 3 \cdot \log_a(u)$

  r is a number

- $a^{\log_a u} = u$

- $a^{(\log_a u)}$

- $\log_a(1) = 0$, from the graph.

- $e^{\ln x} = x$

$1 = a^0$, $\log_a(a^0) = 0$

- $\log_a(a^1) = 1$

(a) $\ln(e^5)$

$= \log_e(e^5) = 5$

(b) $4^{\log_4 3 + \log_4 2} = 4^{\log_4(3 \cdot 2)} = 6$

c) $\log_7(1) = \log_7(7^0) = 0$

$③ \log_a(x^r) = r \cdot \log_a(x)$

Expand and Condense

Assume $x, y, z > 0$. Expand the first expression as far as possible; condense the second into one logarithm.

(a) $\ln\left(\frac{x^3\sqrt{y}}{z}\right)$ (b) $2\log_3 x - \frac{1}{2}\log_3 y.$

Name the property used at each step.

① $\log_a(u \cdot v) = \log_a u + \log_a v$ ② $\log_a\left(\frac{u}{v}\right) = \log_a u - \log_a v$

(a) $\ln\left(\frac{x^3\sqrt{y}}{z}\right) = \ln(x^3 \cdot \sqrt{y}) - \ln(z)$

$= \ln(x^3) + \ln(y^{\frac{1}{2}}) - \ln(z)$

$= 3 \cdot \ln(x) + \frac{1}{2} \cdot \ln(y) - \ln(z)$

(b) $2 \cdot \log_3(x) - \frac{1}{2} \cdot \log_3(y)$

$= \log_3(x^2) - \log_3(\sqrt{y})$

$= \log_3\left(\frac{x^2}{\sqrt{y}}\right)$

Solve, Then Check the Original Domain

Solve over the real numbers:

$$\log_3 x + \log_3(x - 24) = 4.$$

Write the domain restrictions first. Combine the logarithms, solve the resulting equation, and ~~check every candidate~~.

$\log_3(x\cdot(x-24)) = 4$

$x-24$: >0

$\log_a(x) = y \iff a^y = x$

$x(x-24) > 0$

$3^4 = (x\cdot(x-24))$

$x>0,\ x-24>0$

$81 = x^2 - 24x$

$x>24$

$x^2 - 24x - 81 = 0$

use "ac" skill to factor.

$(x-27)(x+3) = 0$

$x=27,\ x=-3$

$$\log_3(x) + \log_3(x-24) = 4$$

$$\log_3(27) + \log_3(27-24) = 4$$

$$\log_3(3^3) + 2 = 4$$

$$3 + 2 = 4$$

$$4 = 4$$